Democracy on Rugged Landscapes: Phase Transitions in Optimal Voting Rules

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Abstract

Laws and institutions shape individual outcomes through complex interactions with citizens' diverse circumstances, yet how different voting methods navigate this coupled landscape remains poorly understood. We model collective governance as optimization on NK fitness landscapes, where shared bits (laws) are updated by voting while individual bits (personal traits) remain fixed. A cross-dependency parameter α controls how legislation's effects depend on individual circumstances. We compare eight standard voting methods and a generalized scoring family across landscape ruggedness $K \in \{1, \ldots, 20\}$ and $\alpha \in [0, 1]$ with 1000 runs per configuration.

We find that the optimal voting method undergoes sharp phase transitions as a function of landscape complexity: total score voting dominates on smooth landscapes, a generalized scoring rule with $p = 0.35$ at low-to-moderate ruggedness, Borda count across a wide middle range, and STAR voting at the highest complexity. A two-parameter empirical formula reduces the (K, α) plane to a single complexity axis for visualization. Borda count emerges as uniquely robust, achieving the highest mean fitness and lowest variance across most of the parameter space.

We further introduce a representative democracy model parameterized by identity weight β and candidate selfishness p_{self}. Representation reshapes the complexity-dependent structure even under favorable conditions: total score voting dominates across most regimes, and as identity voting increases, plurality performs surprisingly well while sophisticated methods lose their advantage.

Introduction

Democratic governance requires a population of individuals with diverse needs and circumstances to make binding collective decisions (laws) that affect everyone. A tax policy, zoning regulation, or healthcare mandate interacts differently with each citizen depending on their occupation, health, location, and other personal attributes. This interaction between shared institutions and individual heterogeneity is a defining feature of law as a complex adaptive system (Ruhl, 1996), yet formal models of voting rarely account for it.

We propose a framework that captures this structure directly. Each individual is represented as a binary string on an NK fitness landscape (Kauffman, 1993), partitioned into a shared portion (bits modified collectively through voting, representing laws) and an individual portion (bits that are fixed and unique to each person, representing personal traits. The NK epistatic structure couples these portions: the fitness contribution of each law-bit depends on nearby trait-bits, and vice versa. A cross-dependency parameter α controls the fraction of each bit's dependencies that cross between portions. At $\alpha = 0$, dependencies are purely internal: laws interact only with other laws, so legislation affects all citizens identically. At $\alpha = 1$, dependencies are purely cross-portion: every law's effect depends on individual traits, but traits lack complex internal interactions. Intermediate values produce the richest landscapes; at $\alpha \approx 0.5$, both internal and cross-dependencies are present, creating the highest effective complexity. This parameter formalizes a spectrum familiar from political economy: from purely public goods ($\alpha = 0$) to policies whose effect is entirely mediated by personal circumstances ($\alpha = 1$).

Within this framework, each round of voting is a step of collective optimization on a rugged landscape. Different voting methods: plurality, approval voting, score voting, Borda count, instant-runoff voting (IRV), STAR voting, minimax (Condorcet), and random dictator, aggregate individual preferences differently, and thus navigate the landscape differently over many rounds.

This perspective suggests a useful unifying interpretation. In our model, all voting methods receive the same input: a utility matrix $U \in \mathbb{R}^{M \times 2^V}$ whose (i, p) entry is the fitness voter i would obtain under proposal p. A voting rule can therefore be viewed as a transformation of the same utility field: different methods preserve, compress, or discard different aspects of U before selecting a winner. For example, plurality keeps

only each voter's top choice, score voting sums raw cardinal values, and Borda discards magnitudes but preserves full within-voter ordinal structure. From this viewpoint, voting methods differ not only in normative properties but also in the kind of information processing they perform.

Beyond direct democracy, we develop a unified model of representative democracy grounded in two long-standing debates in political theory. The first is the trustee–delegate debate: does a representative exercise independent judgment (Burke's trustee (Burke, 1774)) or faithfully implement constituent preferences (the delegate model)? We parameterize this as $p_{\text{self}} \in [0, 1]$, the probability a candidate's platform reflects their own interest rather than their constituency's collective welfare. The second is the question of identity versus policy voting: do citizens elect representatives who share their background and circumstances (descriptive representation (Mansbridge, 1999)) or those whose platforms promise the best outcomes? We parameterize this as $\beta \in [0, 1]$, the weight on identity-based versus policy-based candidate evaluation. Candidates' constituencies are defined by nearest-neighbor Hamming distance in individual traits. Those who share your circumstances are your representatives. The cross-dependency α then plays a natural role: when $\alpha = 0$, individual traits are irrelevant to policy outcomes and identity voting carries no information; when $\alpha = 1$, traits fully mediate policy effects, making identity a strong proxy for policy alignment.

We study two core questions. First, how does voting method choice affect long-run collective welfare (mean fitness, variance, and distributional outcomes) across varying landscape ruggedness K and law-trait coupling α? Second, how does the introduction of representation, and the choice of β and p_{self}, alter these dynamics, and in what parameter regimes does representation help or hurt?

Our main finding is that voting methods undergo sharp phase transitions as a function of landscape complexity: total score voting dominates on smooth landscapes, a generalized scoring rule with $p = 0.35$ at low-to-moderate ruggedness, Borda count across a wide middle range, and STAR voting at the highest complexity. These transitions are well-described by a two-parameter empirical formula that reduces the (K, α) plane to a single complexity axis for visualization. Borda count is the most consistent method, achieving both the highest mean fitness and lowest variance across the majority of the parameter space. Under representative democracy, the complexity-dependent regime structure is reshaped even under favorable conditions: total score voting dominates across most regimes, and as identity voting increases, plurality performs surprisingly well while sophisticated methods lose their advantage. Conceptually, these results also suggest a new research question: whether voting rules can be ordered by computable properties of the utility transformation they implement, and whether such properties predict performance across environments.

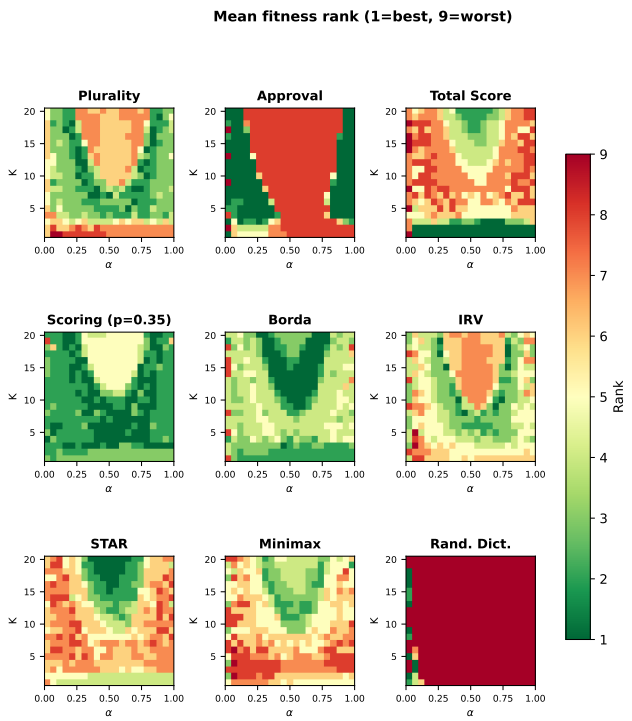

Figure 1: Mean fitness rank (1=best, 9=worst) for each voting method across the full (K, α) grid. Green indicates top performance; red indicates poor performance. Regime transitions are visible as sharp boundaries between green and red regions.

Related Work

NK landscapes and collective search. The NK landscape model (Kauffman, 1993) provides a tunable framework for studying optimization on rugged fitness landscapes, parameterized by the number of components N and the degree of epistasis K. It has been widely applied to organizational decision-making (Levinthal, 1997; Rivkin, 2000), where groups search for high-fitness configurations under varying degrees of interdependence. In these models, organizations navigate rugged landscapes through hierarchical decomposition, imitation, or parallel search. Our work extends this tradition by modeling the search process as democratic voting rather than managerial or evolutionary search, and by introducing a partition between shared (law-like) and individual (trait-like) components

that creates voter heterogeneity absent from prior NK collective search models.

Computational social choice. The formal study of voting methods has a long history (Arrow, 1951; Gibbard, 1973; Satterthwaite, 1975). Arrow's impossibility theorem established that no rank-order voting system can satisfy a small set of reasonable criteria simultaneously, while the Gibbard–Satterthwaite theorem showed that deterministic, non-dictatorial methods are susceptible to strategic manipulation. Computational approaches have since analyzed the complexity, manipulability, and axiomatic properties of various methods (Brandt et al., 2016). However, most analyses focus on single-shot elections or equilibrium outcomes rather than iterated collective optimization over many rounds, which is the regime most relevant to ongoing governance. Our framework bridges this gap by evaluating voting methods as long-run optimizers on a landscape where the quality of collective decisions accumulates over time.

Law as a complex adaptive system. Legal scholars have increasingly applied complexity theory to law (Ruhl, 1996, 2008), arguing that legal systems exhibit emergent behavior, path dependence, and co-evolution with the populations they govern. Our model operationalizes this perspective: laws (shared bits) and individual traits co-determine fitness through epistatic interactions, and the voting process drives adaptation of the legal component. The cross-dependency parameter α formalizes the degree to which legislation interacts with individual circumstances, a central concern in regulatory design that complexity-theoretic analyses of law have identified but not previously modeled at the level of voting mechanisms.

Model

We model a population of M individuals making collective decisions on an NK fitness landscape. Each individual i is represented by a binary string $\mathbf{x}_i \in \{0,1\}^N$, partitioned into a shared portion $\mathbf{s} \in \{0,1\}^{N_s}$ (identical across all individuals, representing laws) and an individual portion $\mathbf{t}_i \in \{0,1\}^{N_t}$ (unique and fixed, representing personal traits), where $N_s + N_t = N$.

Fitness Landscape

Fitness is computed according to the standard NK model (Kauffman, 1993). Each bit position j contributes a fitness value f_j that depends on the bit at position j and K other positions determined by a dependency matrix D:

$$F(\mathbf{x}_i) = \sum_{j=1}^{N} f_j(\mathbf{x}_i[j],\ \mathbf{x}_i[d_{j,1}], \ldots, \mathbf{x}_i[d_{j,K}])\,, \qquad (1)$$

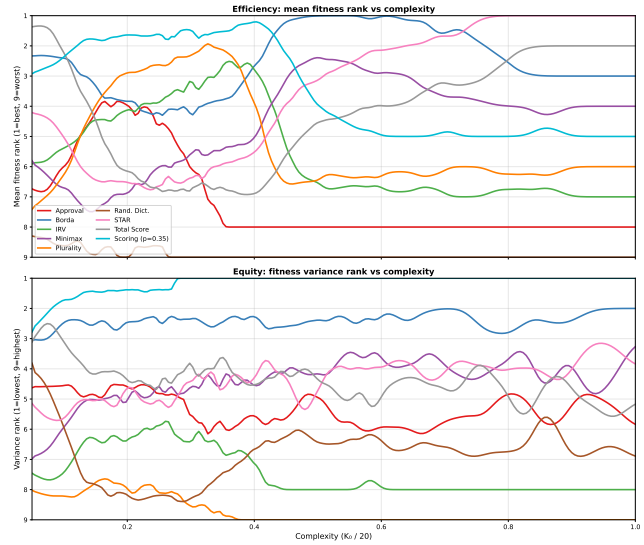

Figure 2: Mean fitness rank (top) and variance rank (bottom) vs. fitted complexity $K_0/20$. The progression Total Score $\rightarrow$ Scoring (p=0.35) $\rightarrow$ Borda $\rightarrow$ STAR is visible for efficiency; both Borda and Scoring (p=0.35) dominate equity.

where $d_{j,1}, \ldots, d_{j,K}$ are the K positions on which position j depends. Each f_j maps a $(K+1)$-bit substring to a fitness value drawn uniformly from $[-1, 1]$. Increasing K increases the ruggedness of the landscape, creating more local optima and making optimization harder.

Cross-Dependency Parameter α

We control the coupling between shared and individual portions through a cross-dependency fraction $\alpha \in [0, 1]$. For each bit position j, a fraction α of its K dependencies are drawn from the other portion, while the remaining $1 - \alpha$ are drawn from the same portion. At $\alpha = 0$, dependencies are entirely internal: laws affect all citizens identically. At $\alpha = 1$, dependencies are entirely cross-portion: individual traits mediate all policy effects but lack complex internal structure. At $\alpha \approx 0.5$, both are present, creating the highest effective complexity.

Voting Process

At each time step, the system selects V bit positions uniformly at random from the shared portion (we use $V = 2$ in direct democracy unless otherwise noted). This defines 2^V possible proposals (all binary configurations of those positions). Each individual i evaluates all proposals by computing the fitness they would obtain under each one, producing a utility matrix $U \in \mathbb{R}^{M \times 2^V}$.

A voting method $\mathcal{V}$ then acts on this matrix to select a winning proposal, which is applied to all individu-

als' shared portions. Equivalently, we may view each method as a function

$$\mathcal{V} : \mathbb{R}^{M \times 2^V} \to \{1, \ldots, 2^V\}, \tag{2}$$

or as a two-stage map that first transforms U into proposal scores and then selects the maximiser. Under this interpretation, all methods solve the same task from the same input, but they differ in which information from U they retain. Some use only each voter's top choice, others use full ordinal rankings, others use cardinal magnitudes, and pairwise methods reprocess the same profile through head-to-head comparisons. We study eight standard methods plus one member of a generalized scoring family:

- Plurality: Each voter casts one vote for their highest-utility proposal; the proposal with the most votes wins.
- Approval: Each voter approves all proposals that improve their fitness over the status quo (or all tied-for-best if none improve); the most approved proposal wins.
- Score: Sum each voter's raw utility for each proposal; highest total wins.
- Borda: Each voter ranks proposals; points are assigned by rank and summed.
- Instant-Runoff (IRV): Iteratively eliminate the proposal with the fewest first-place votes, redistributing those votes, until a majority winner emerges.
- STAR: Sum scores to find the top two proposals, then elect the one preferred head-to-head by more voters.
- Minimax (Condorcet): Select the proposal whose worst pairwise margin of defeat is minimized.
- Random Dictator: A uniformly random voter's top choice is selected. Serves as a baseline.
- Scoring ($p = 0.35$): A positional scoring rule where rank k (out of n) receives score $((n-1-k)/(n-1))^{0.35}$. This is a member of the family of positional scoring rules (Saari, 1995) that interpolates between antiplurality ($p \to 0$) and plurality ($p \to \infty$), with Borda at $p = 1$. At $p = 0.35$, the rule is flatter than Borda, assigning substantial weight across the ranking while remaining purely ordinal. This exponent was identified from a sweep over the scoring family as the value that dominates at low-to-moderate complexity.

Representative Democracy

In the representative variant, C candidates are drawn uniformly at random from the population. Each candidate c represents a constituency: voters assigned by nearest-neighbor Hamming distance on individual traits.

Platform formation (p_{self}). With probability p_{self}, candidate c acts as a trustee and selects the proposal maximizing their own fitness; with probability $1 - p_{\text{self}}$, they act as a delegate and maximize mean constituency welfare. This interpolates between the classical delegate ($p_{\text{self}} = 0$) and trustee ($p_{\text{self}} = 1$) models.

Candidate election (β). Voters evaluate candidates by a convex combination of policy utility $u_{\text{policy}}(i, c) = F_i(\pi_c)$ and identity utility $u_{\text{id}}(i, c) = 1 - d_H(\mathbf{t}_i, \mathbf{t}_c)/N_t$:

$$u(i, c) = (1 - \beta)\, \tilde{u}_{\text{policy}}(i, c) + \beta\, u_{\text{id}}(i, c), \tag{3}$$

where $\beta \in [0, 1]$ is the identity weight and both components are normalized to $[0, 1]$. The same voting method used in direct democracy then elects a winning candidate whose platform is implemented.

The (β, p_{self}) plane nests four limiting cases. Rows: election type (policy vs. identity) $\times$ platform type (constituency vs. self-interested):

Model	β	p_{self}
Trustee	0	1
Delegate	0	0
Identity-trustee	1	1
Identity-delegate	1	0

The cross-dependency α interacts with β: when $\alpha = 0$, identity utility carries no policy information and $\beta > 0$ introduces pure noise; when $\alpha = 1$, traits fully mediate policy effects, making identity a strong proxy for policy alignment.

Experiments

All experiments use $N = 50$ bit positions with a voting portion of 0.5 ($N_s = N_t = 25$), a population of $M = 100$ individuals, $V = 2$ bits voted on per round, and all individuals sharing identical initial voting bits with unique, randomly generated individual bits.

Experiment 1: $K \times \alpha$ Sweep (Direct Democracy)

We sweep landscape ruggedness $K \in \{1, 2, \ldots, 20\}$ and cross-dependency $\alpha \in \{0.00, 0.05, 0.10, \ldots, 1.00\}$ (21 levels) under direct democracy for all eight standard voting methods. Each configuration is run for $150+50K$ iterations with 1000 independent runs, providing high-resolution coverage of the full (K, α) plane. For each

run we record mean fitness, fitness variance, minimum and maximum fitness at every iteration.

In addition, we sweep a generalized positional scoring family parameterized by exponent p: rank k (out of n) receives score $((n-1-k)/(n-1))^p$. This family includes antiplurality ($p \to 0$), Borda ($p = 1$), and plurality ($p \to \infty$) as special cases. We evaluate 28 exponents spanning $p \in [0.05, 10]$ across the full (K, α) grid with 1000 runs per configuration, identifying $p = 0.35$ as the exponent that dominates at low-to-moderate complexity. This ninth method is included in all subsequent analyses.

Experiment 2: Representative Democracy

Using the representative democracy model described in Section 3, we sweep $\beta \in \{0, 0.5, 1\}$ and $p_{\text{self}} \in \{0, 0.5, 1\}$ (9 configurations) with $C = 5$ candidates selected uniformly at random, $V = 4$ bits voted on per round (16 proposals), across the full landscape grid: $K \in \{1, \ldots, 20\}$ and $\alpha \in \{0.0, 0.1, \ldots, 1.0\}$. All nine voting methods are tested, yielding 17,820 configurations with 500 independent runs each. The larger proposal space ($V = 4$ versus $V = 2$ in Experiment 1) ensures that candidates can hold distinct platforms and constituency welfare can meaningfully diverge from population welfare. Using the same (K, α) grid as Experiment 1 allows direct comparison of the fitted complexity axis under representation.

Results

Direct Democracy: $K \times \alpha$ Sweep

We present three main findings from the full (K, α) sweep: (1) method-specific phase diagrams revealing regime-dependent performance, (2) a fitted complexity parameter that aids visualization by reducing the (K, α) plane to a single axis, and (3) a dissociation between efficiency (mean fitness) and equity (fitness variance).

Mean Fitness Rank Phase Diagram Figure 1 shows the rank of each voting method (1=best, 9=worst) by terminal mean fitness at every (K, α) configuration.

Regime transitions. No single method dominates. Instead, the (K, α) plane is partitioned into regimes where different methods are optimal. At the lowest complexity ($K \leq 3$, $\alpha > 0$), total score voting ranks first: a simple utilitarian sum suffices when the landscape is nearly additive. Among the eight classical methods, as K increases to 4–9, plurality takes over in the mid-α range. However, the generalized scoring rule with $p = 0.35$ (identified from a comprehensive sweep over exponents in $[0.05, 10]$) dominates this entire low-to-moderate regime, outperforming plurality while maintaining Borda's favorable equity properties (Figure 2). At moderate complexity ($K \approx 8$–14), Borda count dominates: full ordinal information becomes essential on a rugged landscape. At the highest complexity ($K \geq 15$), STAR voting emerges as rank 1 in the mid-α region, with its two-stage mechanism (scoring followed by pairwise runoff) uniquely suited to maximally rugged, conflicted landscapes.

Approval voting's phase transition. Approval voting exhibits a U-shaped pattern: it ranks first at extreme α values ($\alpha \approx 0$ and $\alpha \approx 1$) but drops to rank 7 across the entire mid-α band ($0.25 \leq \alpha \leq 0.80$) for $K \geq 5$. This boundary is sharp, not gradual, and reflects a structural failure: approval conditions on the status quo, which becomes a poor proxy for the collective optimum when individual circumstances partially mediate policy effects.

Random dictator. Random dictator is rank 9 (worst) at nearly every configuration except $\alpha = 0$, confirming its role as a coordination-free baseline.

A Fitted Complexity Parameter To visualize regime transitions on a single axis, we fit parabolic iso-complexity curves

$$K = a\,(K_0 - 1)^2 \left(\alpha - \tfrac{1}{2} - \frac{b}{K_0 - 1}\right)^2 + K_0, \qquad (4)$$

where $K_0 \in [1, 20]$ indexes complexity and a, b are fitted by maximizing rank-vector uniformity along iso-K_0 contours (Gaussian-kernel-weighted rank variance, bandwidth $\varepsilon = 0.5$). The fitted values are $a = 2.35$, $b = 0.29$. The center $\mu(K_0) = \frac{1}{2} + b/(K_0 - 1)$ converges to $\frac{1}{2}$ at high complexity, consistent with $\alpha = \frac{1}{2}$ producing the richest dependency structure. For $K_0 = 1$, where the parameterization is singular, we treat the degenerate iso-complexity contour separately as the flat line $K = 1$.

Efficiency and Equity vs. Complexity Figure 2 shows mean fitness rank (top) and variance rank (bottom) for all nine methods plotted against normalized complexity $K_0/20$. The regime transitions collapse cleanly onto this single axis for both measures.

Efficiency. Total score dominates at low complexity, then scoring ($p = 0.35$) takes over through the low-to-moderate range, Borda dominates the mid-range, and STAR emerges at the highest complexity. Among classical methods alone, plurality briefly leads in the low-to-moderate range, but $p = 0.35$ strictly dominates it there.

Equity. Both Borda and scoring ($p = 0.35$) achieve rank 1–2 for variance (most equitable) across nearly the entire complexity range. Scoring ($p = 0.35$) achieves the best or near-best equity while simultaneously dominating efficiency at low-to-moderate complexity, a combi-

nation no classical method achieves. Plurality and random dictator consistently produce high variance; IRV shows poor equity at high complexity.

Representative Democracy

We sweep $\beta \in \{0, 0.5, 1\}$ and $p_{\text{self}} \in \{0, 0.5, 1\}$ across the full (K, α) grid ($K \in \{1, \ldots, 20\}$, $\alpha \in \{0.0, 0.1, \ldots, 1.0\}$) with $V = 4$ bits, $C = 5$ candidates, and 500 runs per configuration.

The (β, p_{self}) Grid Figure 3 presents a 3×3 grid of rank-versus-complexity curves, one panel for each (β, p_{self}) configuration, using the same fitted complexity axis (Eq. 4) for comparability with direct democracy.

Preservation of regime structure. At ($\beta = 0$, $p_{\text{self}} = 0$) (policy-based voting with altruistic delegates) the rank-versus-complexity curves most closely resemble direct democracy. Multiple methods compete across the complexity axis, and the progression from simple to sophisticated methods is visible. This configuration represents ideal representation: voters evaluate candidates on policy merit, and candidates faithfully serve their constituencies.

Collapse under selfishness. As p_{self} increases (moving rightward in the grid), the rank curves become increasingly chaotic. At $p_{\text{self}} = 1$, no method clearly dominates at any complexity level; the rank curves cross repeatedly and erratically, with no stable ordering. Self-interested representatives introduce noise that overwhelms the complexity-dependent signal: the proposals implemented reflect individual self-interest rather than any aggregation of collective preferences, rendering the choice of voting method effectively random in its consequences.

Collapse under identity voting. As β increases (moving downward), method differentiation also decreases, though less severely than under high p_{self}. At $\beta = 1$, voters elect representatives based on shared traits rather than policy outcomes. The resulting representatives may share their constituents' circumstances but do not necessarily champion their preferred policies, reducing the signal available to the voting method.

The worst case. The ($\beta = 1$, $p_{\text{self}} = 1$) corner (identity voting with self-interested trustees) produces the least differentiated rank curves. Here, representatives are elected for who they are, not what they propose, and then implement whatever benefits themselves. The voting method becomes largely irrelevant: it cannot compensate for a representation layer that discards both policy information (high β) and constituency alignment (high p_{self}).

Delegate Advantage The delegate model ($p_{\text{self}} = 0$) consistently outperforms the trustee model ($p_{\text{self}} = 1$) in mean fitness when $\alpha > 0$, with the largest gaps at mid-α and high K. At $K = 10$, $\alpha = 0.5$, the delegate advantage is 1.39 mean fitness units (best-method fitness of 3.45 vs. 2.04). At $K = 20$, $\alpha = 0.5$, the delegate model more than doubles collective welfare (1.82 vs. 0.78). At $\alpha = 0$, the advantage vanishes entirely; on landscapes with no cross-dependency, self-interest and constituency welfare coincide.

The β–α Interaction Identity-based voting ($\beta = 1$) underperforms policy-based voting ($\beta = 0$) most severely at mid-α. At $K = 10$, $\alpha = 0.5$, the penalty is 1.57 fitness units (best-method fitness of 3.45 under $\beta = 0$ vs. 1.88 under $\beta = 1$). At $\alpha = 0$, the three β values converge exactly: individual traits carry no policy information, so identity voting is pure noise but harmless noise. At $\alpha = 1$, the β values nearly converge: traits fully mediate policy effects, making identity similarity a reasonable proxy for policy alignment.

Discussion

Direct Democracy: Complexity-Dependent Optimization

Under direct democracy, no voting method universally dominates: the optimal method depends on the complexity of the underlying decision landscape. This goes beyond the classical impossibility theorems of Arrow and Gibbard-Satterthwaite, which characterize methods by axiomatic properties on fixed preference profiles. Our framework reveals that the empirical performance ordering of methods is a function of landscape complexity and undergoes sharp phase transitions.

The progression Total Score $\rightarrow$ Scoring ($p = 0.35$) $\rightarrow$ Borda $\rightarrow$ STAR has a natural interpretation in terms of information aggregation depth. On smooth landscapes, preferences are nearly aligned and a simple sum of utilities suffices. As ruggedness increases, preferences fragment: a flatter ordinal scoring rule ($p = 0.35$) that spreads weight across the ranking, outperforms both the raw sum and plurality's hard cutoff. At moderate complexity, full ordinal information becomes essential (Borda, $p = 1$), and at the highest complexity, even ordinal information must be combined with pairwise verification (STAR). Each successive method in the progression extracts richer information from voters' preferences, at increasing computational and communicative cost. Among classical methods plurality occupies part of the regime where $p = 0.35$ dominates, suggesting that plurality captures a real low-complexity signal but that its hard cutoff is too coarse to exploit it fully.

Voting Rules as Utility Transforms

In our model, all voting methods operate on the same utility matrix $U \in \mathbb{R}^{M \times n}$ and can be understood as different transformations of this common input: plurality

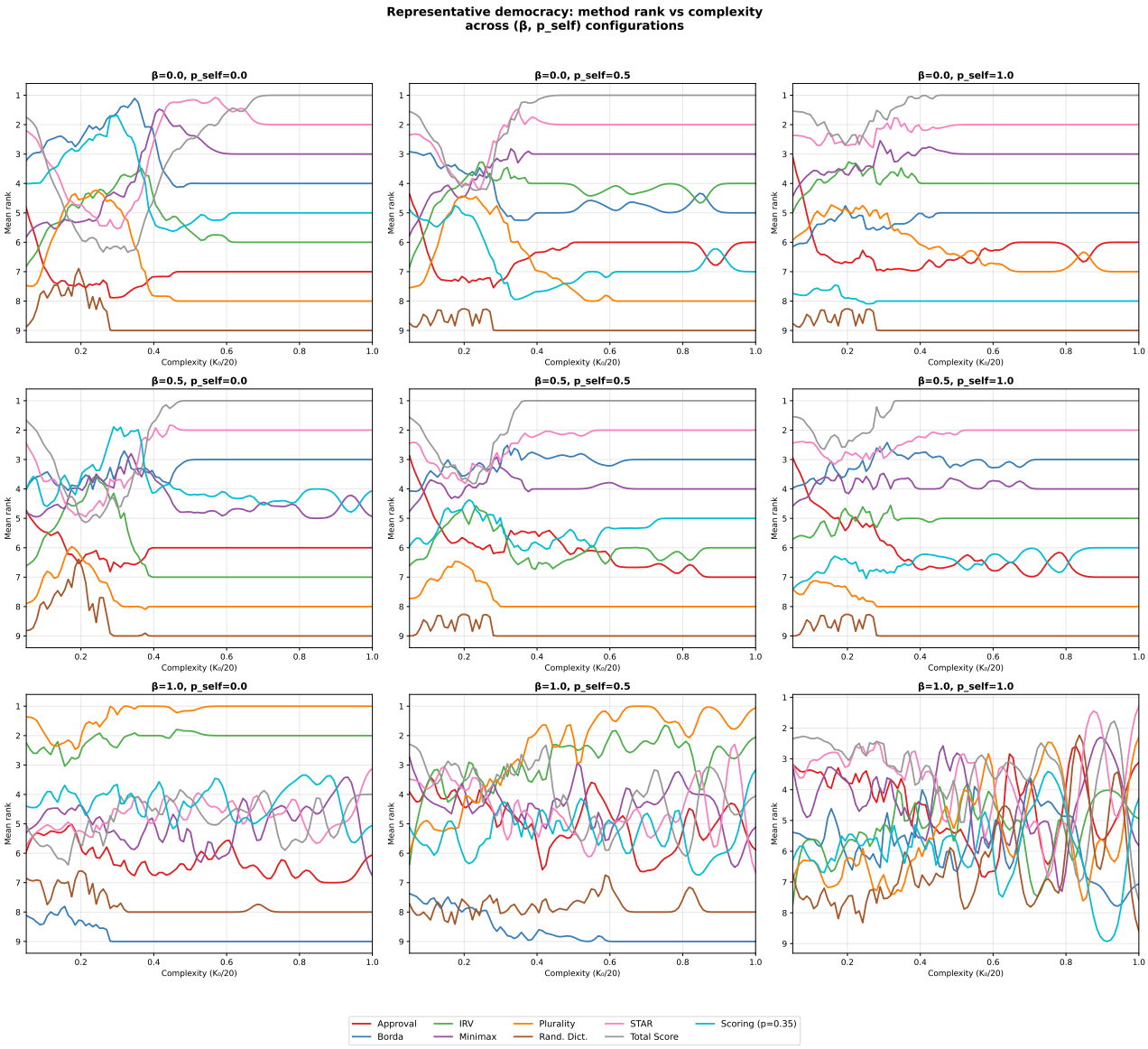

Figure 3: Mean rank vs. complexity for all nine methods across the 3×3 (β, p_{self}) grid. Rows vary β (identity weight: 0, 0.5, 1); columns vary p_{self} (candidate selfishness: 0, 0.5, 1). Representation reshapes the regime structure: high p_{self} produces chaotic rankings; high β favors simpler methods.

compresses each row to an argmax, Borda replaces magnitudes with ordinal ranks, score voting sums cardinal values directly, and pairwise methods reprocess the profile through head-to-head comparisons. Each transformation preserves different features of U—top-choice information, ordinal structure, cardinal intensity, or pairwise margins—while discarding others.

This framing suggests a theoretical research program. Given assumptions about the fitness landscape (ruggedness, cross-dependency structure), one might prove which properties of the utility transform lead to optimal collective optimization, connecting computational social choice to optimization theory. For instance, our results suggest that on smooth landscapes, preserving cardinal magnitudes is optimal (score voting), while on rugged landscapes, preserving full ordinal structure (Borda) or pairwise comparisons (STAR) becomes necessary. A formal theory characterizing this relationship could yield optimality results for voting methods as a function of landscape properties, rather than relying on axiomatic criteria alone.

Borda's Robustness and Equity

Borda count's simultaneous dominance in both mean fitness and variance across a wide complexity range is the most practically relevant finding. Borda is never catastrophically bad (worst observed rank ≈ 4) and is the only method that achieves both high efficiency and high equity across the majority of the parameter space. This complements classical axiomatic work that identifies Borda by distinctive structural properties (Young, 1974), but in a dynamic, multi-round optimization setting far from the one-shot environments usually studied in social choice theory.

Representation as an Information Bottleneck

Representation reshapes the regime structure even under favorable conditions. At ($\beta = 0$, $p_{\text{self}} = 0$), the direct democracy ordering is partially visible but already altered: representation acts as a lossy compression, and even faithful delegates cannot fully reconstruct the population-level signal that direct democracy exploits. The two representation parameters introduce qualitatively different distortions: high p_{self} produces chaotic rank curves with no stable method ordering, while high β favors simpler methods; plurality performs surprisingly well when voters select on identity rather than policy, as its concentrated signal is more resistant to noise than methods requiring richer preference information. The implication for institutional design is that voting method choice matters most under faithful representation; as either β or p_{self} increases, the binding constraint shifts from aggregation quality to representation quality.

Limitations

Several simplifications limit generalizability. All agents vote sincerely; strategic voting could qualitatively change method comparisons, particularly for IRV and Borda (Gibbard, 1973). Individual portions are fixed; in reality, personal circumstances respond to collective decisions. The NK landscape is random; results are averages over realizations, not predictions for specific policy domains. The proposal space ($V = 2$ for direct, $V = 4$ for representative) is small. Our constituency assignment uses Hamming distance, a simplification of real representational geography. Finally, we study only three discrete values of β and p_{self}; finer resolution might reveal additional structure.

Conclusion

We have presented an NK landscape framework for studying voting methods as collective optimizers. The main findings are:

1. Complexity-dependent method ordering. Under direct democracy, method performance undergoes sharp phase transitions: total score $\rightarrow$ scoring ($p = 0.35$) $\rightarrow$ Borda $\rightarrow$ STAR as complexity increases. No method universally dominates.

2. Borda's robustness. Borda count is simultaneously the most efficient and most equitable method across a wide complexity range. A sweep over the generalized scoring family reveals that $p = 0.35$ (a flatter ordinal scoring rule) dominates plurality in the low-to-moderate regime while matching Borda's favorable variance properties.

3. Voting rules as utility transforms. The model frames voting methods as transformations of a common utility matrix, suggesting that formal connections between transform properties and landscape-dependent optimality may be possible.

4. Representation as information bottleneck. Under representative democracy, the regime structure is reshaped by candidate selfishness and identity voting, with simpler methods gaining advantage as representation degrades.

These results suggest that debates over voting reform are implicitly debates about the complexity of the policy landscape. Future work might explore strategic voting, develop formal connections between utility transform properties and landscape-dependent optimality, introduce fitness-biased candidate selection to model incumbency advantage, and extend the framework with a third, socially transmitted "belief" portion of the bitstring so that opinion dynamics on networks can interact with collective optimization.

Acknowledgments

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